Department of Mathematics and Statistics
Texas Tech University

Exam II

Fall 2026
Practice Exam (Version B)

Name (print): _____ **R-Number:** _____ **Seat Number:** _____

- Please read the directions for all problems carefully.
- Show all solutions in the space provided. If necessary, the back of a page may be used to complete an answer.
- Show a sufficient and reasonable amount of work to justify your answers.
- Please circle your final answer.
- Use of any electronic device during the examination is strictly prohibited. Please keep smartphones, smartwatches, and calculators out of view.

My signature below demonstrates that I will comply with the directions above, the work on this exam is my own, and I did not receive or give assistance on the exam.

Signature: _____

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1. (10 pts) Consider the function $f(x) = x^3 - 4x + 1$.

(a) (6 pts) Use the definition of the derivative to show that $f'(x) = 3x^2 - 4$.

(b) (4 pts) Use part (a) to find $f'(3)$ and $f''(x)$.

2. (10 pts) Use the definition of the derivative to show that $f'(x) = \dfrac{1}{2\sqrt{x}}$ if $f(x) = \sqrt{x}$.

3. (30 pts) Determine the derivative of the following functions. (3 pts each)

(a) $f(x) = 7x^4 - 2x^2 + 9x - 3$

(b) $g(x) = \sqrt[4]{x^3}$

(c) $h(x) = \dfrac{2x^5 - x + 4}{x^2}$

(d) $f(x) = x^5 \cos(x)$ (Do not simplify.)

(e) $y = \dfrac{\ln(x)}{x^4}$ (Do not simplify.)

(f) $s(t) = \dfrac{1}{\cos(t)}$

(g) $f(x) = e^{(2x^2+5x-1)}$

(h) $r(\theta) = \big[\ln(\sin\theta)\big]^2$

(i) $f(x) = \ln\big(\sqrt{x^6 + x^2 + 7}\big)$

(j) $y = \sqrt{\dfrac{x^2+1}{x^3-5}}$ (Do not simplify.)

4. (10 pts) Consider the curve defined implicitly by $x^2 + xy + y^2 = 7$.

(a) (6 pts) Use implicit differentiation to find $\dfrac{dy}{dx}$.

(b) (4 pts) Verify that $(1, 2)$ lies on the curve, then evaluate $\dfrac{dy}{dx}$ at $(1, 2)$.

Total: _____ / 60